REPRODUCING MEANFLOW: A REPLICATION STUDY OF ONE-STEP GENERATIVE MODELING

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ABSTRACT

We study the reproducibility of selected MeanFlow ablations. The completed ImageNet-256 study reproduces Table 1(e) with the authors' JAX implementation, five loss exponents and two training seeds per exponent. At epoch 80, $p = 1$ performs best in both seeds, with EMA FID-50K 61.40 ± 0.06; $p = 2$ gives 97.78 ± 0.24 versus the paper's 69.19. Values are means $\pm$ sample SD ($n = 2$). This is partial numerical replication with limited evidence from two seeds. Results for the group's sampling-ratio and Jacobian-vector-product ablations await integration.

***Index Terms*—** MeanFlow, reproducibility, one-step generation, adaptive loss weighting, Fréchet Inception Distance

Contributions. Simon Winther Albertsen adapted and instrumented the authors' JAX code for GPU execution, prepared and validated the ImageNet VAE latent cache, ran both five-exponent sweeps, evaluated FID and analyzed the results. Hjalte Bjoernstrup reproduced the Jacobian-vector-product ablation in Table 1(b). Said Djafar Said reproduced the $r \neq t$ sampling-ratio ablation in Table 1(a).

1. INTRODUCTION

MeanFlow learns average velocity over a time interval for one-step generation [1]. Its training target uses a Jacobian-vector product. It trains from scratch without a pretrained teacher or distillation.

We target Table 1(a), the ratio of sampling $r \neq t$; Table 1(b), Jacobian-vector-product computation; and Table 1(e), loss metrics. The completed loss-metric study uses the authors' JAX code on ImageNet-1K at 256×256. The other experiments' datasets, implementations and results await integration. Fréchet Inception Distance (FID) assesses generation quality; we disclose departures from the released setup.

2. EXPERIMENT: ADAPTIVE LOSS METRICS

We reproduce MeanFlow's Table 1(e) [1] to test whether adaptive loss weighting improves one-step FID and whether $p = 1$ performs best among $p \in \{0, 0.5, 1, 1.5, 2\}$. We train ten models: two per p, with training seeds 42 and 43. Within each sweep, only `method.norm_p` varies.

Objective and controls. For training example i, let $q_i = \|u_\theta - \mathrm{sg}(u_{\mathrm{tgt}})\|_2^2$, summed over latent elements. The released code minimizes

$$\mathcal{L}_p = \frac{1}{B}\sum_{i=1}^{B} \mathrm{sg}\left[(q_i + c)^{-p}\right] q_i, \qquad c = 0.01, \qquad (1)$$

Here sg stops gradients through the Jacobian-vector-product target and adaptive weight. Setting $p = 0$ gives squared L2 loss; $p > 0$ downweights larger residuals. We use the released `norm_eps`=0.01. Training-loss scales vary with p and are not compared.

We use class-conditional MeanFlow/DiT-B/4 (131.2M parameters), trained from scratch for 80 epochs (400,320 updates) with global batch 256 on one NVIDIA B200 per run. AdamW uses a constant learning rate of 10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.95)$ and zero weight decay; EMA decay is 0.9999. All runs use FP32, logit-normal time sampling $(-0.4, 1.0)$, $r \neq t$ for 25% of examples, $(t, t-r)$ conditioning, class dropout 0.1 and no classifier-free guidance.

Implementation and evaluation. We use the authors' JAX code [2], commit `d70cb55`. We encode ImageNet-1K's 1,281,167 training images, center-cropped to 256×256, with the pretrained VAE `pcuenq/sd-vae-ft-mse-flax` at revision `7581b0a`. We sample $32 \times 32 \times 4$ latents from cached posterior means and standard deviations, with scaling 0.18215. We evaluate EMA FID on 50,000 decoded images every ten epochs with one network evaluation (1-NFE), fixed sampling seed 0, and the authors' JAX Inception/resize pipeline and reference statistics. We report epoch 80, averaging the two measured FIDs without pooling sample statistics.

Table 1. Epoch-80 EMA FID-50K (1-NFE), mean ± sample SD ($n = 2$ training seeds, ddof=1). Δ = paper − our mean. Bold marks minimum FID per column.

p	Paper	Seed 42	Seed 43	Mean ± SD	Δ
0.0	79.75	80.23	81.47	80.85 ± 0.88	−1.10
0.5	63.98	64.60	65.22	64.91 ± 0.44	−0.93
1.0	**61.06**	**61.36**	**61.44**	**61.40** ± 0.06	−0.34
1.5	66.57	67.32	66.62	66.97 ± 0.49	−0.40
2.0	69.19	97.61	97.95	97.78 ± 0.24	−28.59

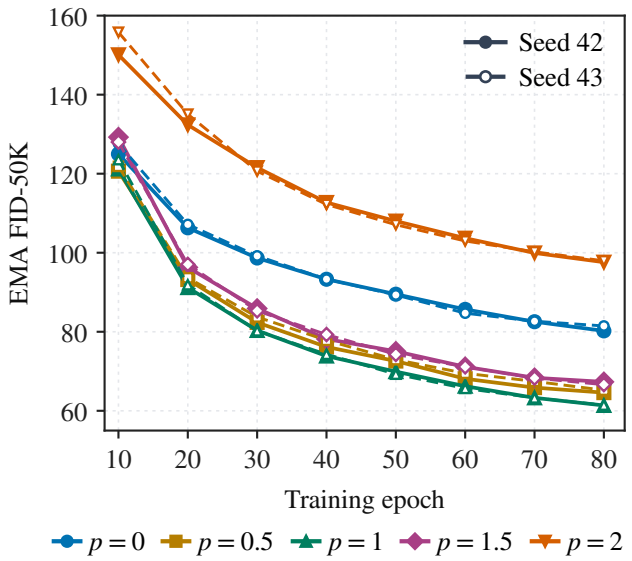


Fig. 1. Recorded EMA FID-50K at ten-epoch checkpoints. Colour identifies p; solid/filled denotes seed 42, dashed/open seed 43. Paired curves largely overlap.

The TPU release pins JAX/JAXlib 0.4.27; our GPU runs use 0.11.2, Python 3.12.3, Flax 0.12.10 and SciPy 1.18.1. Patches adapt JAX/SciPy APIs, single-process startup, local VAE loading and data-loader workers; a wrapper records results and timings. We freeze latent file order; epochs repeat that order without latent flips, as in the released loader. Historical training seeds, file order and stabilizer values are unspecified, preventing bitwise historical replication.

Results. Setting $p = 1$ is best in both seeds, with mean FID 61.40 ± 0.06 (Table 1). For $p \leq 1.5$, individual scores are 0.05 to 1.72 above the paper. For $p = 2$, FIDs are 97.61 and 97.95 versus 69.19; their mean exceeds the paper by 28.59 and performs worse than our $p = 0$ results. Both seeds' FIDs decrease at every checkpoint, with $p = 2$ highest throughout (Figure 1). This is partial numerical replication; the persistent $p = 2$ discrepancy has no identified cause.

Recorded run time totals 63.03 GPU-hours: 31.61 for seed 42 and 31.42 for seed 43. Runs take 6.281 to 6.325 hours; parallel sweeps take 6.33 and 6.29 elapsed hours, respectively. Timings include compilation, I/O, evaluation, checkpointing and recording; preprocessing and idle allocation time are excluded. Allocated/billed time is unavailable. Eight FID evaluations take 21.1 to 25.1 minutes per run; median logged throughput from step 500 is 18.97 to 19.07 updates/s.

After inspecting seed 42, we repeated the full sweep with only the training seed changed to 43 to check sensitivity to training randomness. All ten runs completed successfully. Two seeds provide limited descriptive evidence and cannot establish robustness, significance or outlier status.

3. DISCUSSION

At $p = 2$, the mean FID exceeds the paper by 28.59 ($\Delta = -28.59$), with both seeds worse than squared L2. The observed seed difference is 0.34. Seed 42 is the released default; the historical seed is unspecified. The posthoc second sweep provides limited two-seed evidence.

The $p = 2$ weight, $w = (q + 0.01)^{-2}$, could amplify stabilizer sensitivity. The paper's $c = 10^{-3}$ is illustrative; its historical value is unknown, while ours follows the code's 0.01. GPU/software differences and file order could also contribute. None was isolated experimentally; the cause remains unresolved.

3.1. Limitations

Metric. FID-50K is the paper's main quality metric and our sole measure. It combines fidelity and coverage, making failure diagnosis difficult; precision/recall and human assessment would provide complementary evidence [3].

Models. Table 2 includes Shortcut, IMM, DiT and SiT [1], but no standalone Rectified Flow baseline. Rankings depend on sampling and training budgets; they are not controlled comparisons at matched training compute. Our B/4 ablation cannot verify the larger-model claims.

Datasets. Generation is evaluated on ImageNet-256 and CIFAR-10 [1], leaving other domains and background/noise corruptions untested. VTAB-1k's 19 tasks illustrate broader coverage but measure classification transfer rather than generation [4]. Generalization beyond the reported benchmarks remains unverified.

4. CONCLUSION

For the loss-metric experiment, $p = 1$ gives the lowest FID in both training seeds. Results for $p \leq 1.5$ closely match Table 1(e), while the $p = 2$ discrepancy persists. This is partial numerical replication.

5. REFERENCES

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